BENCHMARKING PARAMETER-EFFICIENT FINE-TUNING: TRADE-OFFS IN ACCURACY, SPEED, AND SPARSITY

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ABSTRACT

Fine-tuning large language models (LLMs) is computationally expensive due to their massive parameter counts. This paper investigates four parameter-efficient fine-tuning (PEFT) methods: LoRA (Low-Rank Adaptation), Sparse LoRA, QLoRA (Quantized LoRA), and High-Rank LoRA (as a proxy for HiRA). We evaluate these methods on three benchmark datasets (IMDB, SST-2, and WikiText-2) using DistilBERT and BERT-base architectures, measuring task accuracy, inference speed, memory footprint, and training efficiency. Key findings indicate that QLoRA achieves the highest classification accuracy (93.28% on IMDB) with 50% memory reduction, but at a 2.5–3.5× training time penalty. Conversely, High-Rank LoRA demonstrates superior language modeling performance (perplexity of 2.62 on WikiText-2), while Sparse LoRA achieves 50% sparsity with minimal accuracy loss, offering a viable path for efficient deployment. Standard LoRA provides the most balanced trade-off across all metrics.

1 INTRODUCTION

Large language models (LLMs) deliver impressive accuracy on many NLP benchmarks, yet their billions of parameters make full fine-tuning prohibitive in terms of memory, wall-clock time, and energy. Updating every weight of a 65B-parameter model can require dozens of high-end GPUs, severely limiting accessibility. Parameter-efficient fine-tuning (PEFT) mitigates this barrier by freezing the model backbone and learning small adapter modules. LoRA and its variants exemplify this approach by injecting low-rank or structured updates that capture task-specific behavior without modifying the full model weights.

This paper benchmarks four adapter families—standard LoRA, Sparse LoRA, QLoRA, and High-Rank LoRA—on three representative tasks (IMDB, SST-2, WikiText-2). We address three practical research questions: (i) How much accuracy is retained compared to full fine-tuning? (ii) What memory and compute savings does each variant offer? (iii) Which method is optimal under constraints such as training speed, deployment latency, or generative quality?

2 LITERATURE REVIEW

We briefly highlight the ingredients behind each adapter variant evaluated in this work.

LoRA freezes the base model and learns a rank-r update $\Delta W = BA$ injected into attention blocks (1). Because $r \ll d, k$, the adapter introduces only $r(d + k)$ parameters, can be merged back into W at inference, and leaves latency unchanged.

Sparse LoRA pushes efficiency further by pruning half of the LoRA weights through magnitude-based schedules and L1 regularization (2). The sparsity pattern is fixed post-training, enabling faster inference once a sparse kernel is available.

QLoRA combines LoRA with 4-bit NF4 quantization, double quantization of scaling factors, and paged optimizers to enable fine-tuning of massive models on a single GPU (3). Only the adapters

remain in higher precision, yielding substantial memory savings at the cost of additional compute overhead.

HiRA relaxes the low-rank assumption by composing two rank-r matrices via Hadamard products, effectively emulating a higher rank without quadratic parameter growth (4). This extra expressiveness particularly helps generative tasks. In our project, we approximate this added expressiveness by replacing HiRA's Hadamard-composed factors with a High-Rank LoRA module, which increases the effective update rank through a structured sum of low-rank components while preserving parameter efficiency.

DistilBERT serves as our compact backbone with 66M parameters (6 transformer layers, 768 hidden dimensions, 12 attention heads) while retaining 97% of BERT's language understanding capability (5). It is distilled from **BERT-base-uncased**, which has 110M parameters (12 transformer layers, 768 hidden dimensions, 12 attention heads). QLoRA uses BERT-base because current 4-bit quantization kernels better support that architecture.

3 METHODOLOGY

We evaluate four PEFT strategies using the Hugging Face `peft` library and PyTorch. All experiments were conducted on the Duke Compute Cluster (DCC).

3.1 MODELS AND DATASETS

We employ **DistilBERT** (66M parameters) as the primary backbone for LoRA, Sparse LoRA, and High-Rank LoRA due to its efficiency. For **QLoRA**, we utilize **BERT-base-uncased** (110M parameters, 12 transformer layers, 768 hidden dimensions, 12 attention heads) because 4-bit quantization support for DistilBERT is limited in current libraries. BERT-base's larger capacity—nearly $1.7\times$ the parameters of DistilBERT—provides a richer representation space, which partly explains QLoRA's classification advantage. While this introduces a model size discrepancy, it reflects real-world usage where QLoRA is typically applied to larger architectures to maximize memory savings from quantization. We explicitly acknowledge this confound and qualify our results accordingly.

We evaluate on three datasets:

- **IMDB**: Binary sentiment classification (25k train / 25k test).
- **SST-2**: Short-text binary sentiment classification from GLUE.
- **WikiText-2**: Causal language modeling to evaluate generative capabilities.

3.2 ADAPTER CONFIGURATIONS

- **LoRA**: Rank $r = 8$, alpha $\alpha = 32$, dropout 0.1. Targets query/value projections.
- **Sparse LoRA**: Identical to LoRA ($r = 8$) but incorporates magnitude pruning during training to induce 50% sparsity in the adapter weights. We employ a *cubic sparsity schedule* where the sparsity ratio s_t at step t follows $s_t = s_f \cdot \left(1 - \left(1 - \frac{t-t_0}{t_f-t_0}\right)^3\right)$ for $t \in [t_0, t_f]$, with $s_f = 0.5$ (final sparsity), t_0 (pruning start), and t_f (pruning end). This gradual ramp-up allows the network to adapt before aggressive pruning. At each pruning step, weights with the smallest magnitudes are zeroed out. We also apply L1 regularization ($\lambda = 10^{-5}$) to encourage weight sparsity throughout training.
- **QLoRA**: Uses 4-bit NormalFloat (NF4) quantization for the base model with double quantization, combined with LoRA adapters ($r = 8$).
- **High-Rank LoRA (HiRA Proxy)**: To test the hypothesis that higher capacity improves complex tasks, we implement a high-rank variant with $r = 32$ and $\alpha = 64$. This serves as a proxy for High-Rank Adaptation (HiRA) to evaluate the impact of increased trainable parameters on capacity-intensive tasks.

3.3 Evaluation Metrics

For classification tasks, we report **Accuracy** and **F1 Score**. For language modeling, we report **Loss** (cross-entropy) and **Perplexity**, computed as $\text{PPL} = \exp(\mathcal{L})$ where $\mathcal{L}$ is the average negative log-likelihood per token. Lower perplexity indicates better predictive performance. Across all tasks, we measure **Training Time** (minutes), **GPU Memory** (MB), **Trainable Parameter %** relative to the base model, and **Throughput** (samples/second) during inference.

4 Results

4.1 Classification Performance (IMDB & SST-2)

Table 1 summarizes the classification results on IMDB. **QLoRA** achieves the highest accuracy (93.28%), effectively compensating for quantization noise through adapter learning. However, this performance comes at significant cost: training time increases by approximately 3× compared to standard LoRA. Sparse LoRA maintains competitive accuracy (within 0.04 percentage points of LoRA) while achieving 50% sparsity, validating the redundancy hypothesis in adapter weights.

Table 1: IMDB Sentiment Classification Results. Note: QLoRA uses BERT-base (110M params); all others use DistilBERT (66M params).

Method	Accuracy	F1	Train %	Time	Memory
LoRA	92.52%	0.9252	1.09%	35.2 min	258 MB
Sparse LoRA	92.48%	0.9248	1.09%	35.2 min	258 MB
QLoRA†	**93.28%**	**0.9328**	**0.40%**	100.8 min	**134 MB**
High-Rank	92.93%	0.9293	1.75%	35.4 min	258 MB

†QLoRA uses BERT-base due to 4-bit quantization library constraints.

Results on SST-2 (Table 2) mirror these findings. QLoRA provides a +2.18 percentage point accuracy improvement but requires significantly longer training. Standard LoRA and High-Rank LoRA offer balanced profiles, completing training in roughly 14 minutes, making them suitable for rapid iterative development.

Table 2: SST-2 Sentiment Classification Results. QLoRA uses BERT-base; others use DistilBERT.

Method	Accuracy	F1	Train %	Time	Memory
LoRA	90.02%	0.9002	1.09%	13.6 min	258 MB
Sparse LoRA	89.22%	0.8922	1.09%	13.6 min	258 MB
QLoRA†	**92.20%**	**0.9220**	**0.40%**	46.9 min	**134 MB**
High-Rank	90.14%	0.9014	1.75%	13.6 min	258 MB

†BERT-base backbone (see Section 3.1).

Figure 1 visualizes the accuracy comparison. The trade-off is clear: quantization boosts accuracy, but this is confounded by the larger base model capacity (BERT-base vs. DistilBERT). While QLoRA achieves the highest absolute performance, it is not Pareto-optimal if training speed is a primary constraint, and its advantage diminishes when normalized for base model size.

4.2 Language Modeling

For language modeling on WikiText-2 (Table 3), the trend reverses. **High-Rank LoRA** achieves the lowest perplexity (2.62), suggesting that generative tasks benefit substantially from increased adapter capacity. QLoRA lags behind with a perplexity of 3.20, indicating that aggressive 4-bit quantization may degrade performance in fine-grained next-token prediction tasks, even if it aids classification.

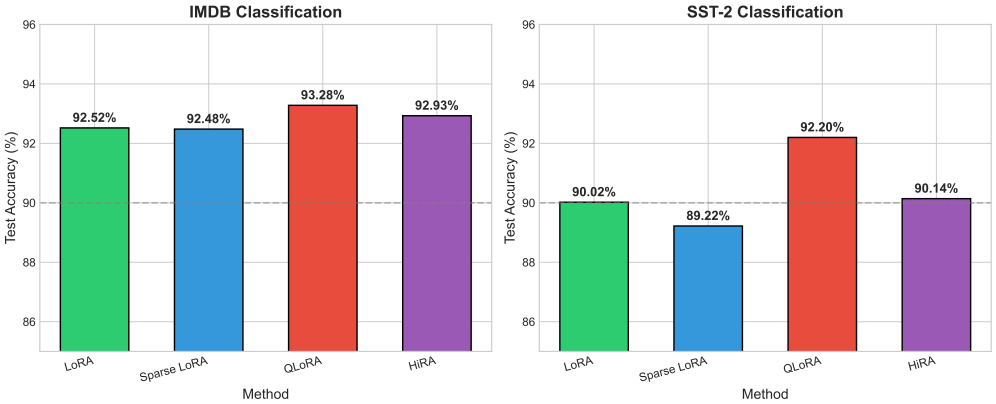


Figure 1: Classification accuracy comparison across IMDB and SST-2 datasets. QLoRA leads in accuracy, while LoRA and High-Rank variants remain competitive with significantly lower computational overhead.

Table 3: WikiText-2 Language Modeling Results. QLoRA uses BERT-base; others use DistilBERT.

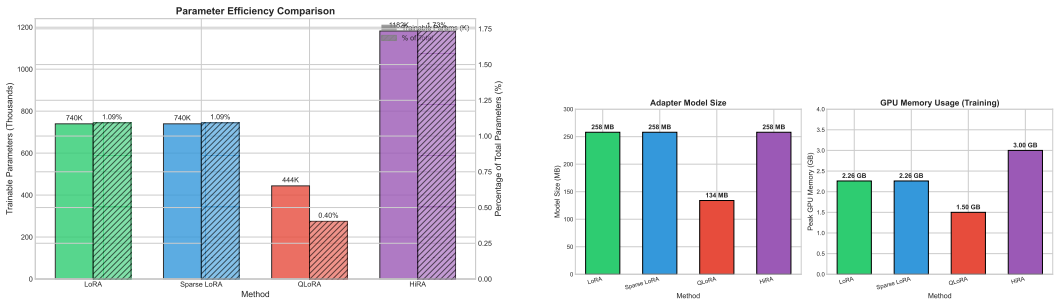
Method	**Loss**	**Perplexity**	**Train %**	**Time**	**Throughput**
LoRA	0.9867	2.68	0.22%	24.1 min	181.7 samples/s
Sparse LoRA	0.9938	2.70	0.22%	24.1 min	∼180 samples/s
QLoRA†	1.1631	3.20	0.40%	60.4 min	100.1 samples/s
High-Rank	**0.9642**	**2.62**	0.87%	24.2 min	169.9 samples/s

†BERT-base backbone (see Section 3.1).

4.3 Efficiency Metrics

Figure 2 compares parameter counts and memory footprint. QLoRA shrinks adapter size by roughly half relative to the other methods, while Sparse LoRA matches LoRA's footprint but unlocks structured pruning. Training-time differences are captured in Figure 3; the hatched QLoRA bars highlight the 2.5–3.5× slowdown despite similar epochs and batch sizes.

Sparsity. Sparse LoRA reaches the target 50% sparsity on every dataset with negligible accuracy drift. The result hints at sizeable redundancy inside low-rank adapters and motivates sparse kernels for inference speedups.



(a) Parameter efficiency

(b) Memory comparison

Figure 2: Efficiency metrics comparison. (a) QLoRA achieves the lowest trainable parameter percentage (0.40%). (b) QLoRA's 4-bit quantization reduces model memory by approximately 48%.

5 Discussion

Task sensitivity. Quantization helps classification yet hurts generation: QLoRA gains +0.76–2.18 percentage points on IMDB/SST-2 but yields 19% worse perplexity on WikiText-2. Consequently,

Figure 3: Training time comparison across all datasets. QLoRA (hatched bars) takes 2.5-3.5× longer than other methods due to quantization overhead.

we recommend QLoRA only for memory-bound classification workloads; High-Rank LoRA or standard LoRA are safer choices for language modeling.

Speed vs. accuracy. Figure 4 shows that QLoRA occupies the high-accuracy corner, even though LoRA and High-Rank LoRA are three times faster to fine-tune. Sparse LoRA tracks the LoRA point despite pruning half its parameters, making it attractive when rapid iteration matters.

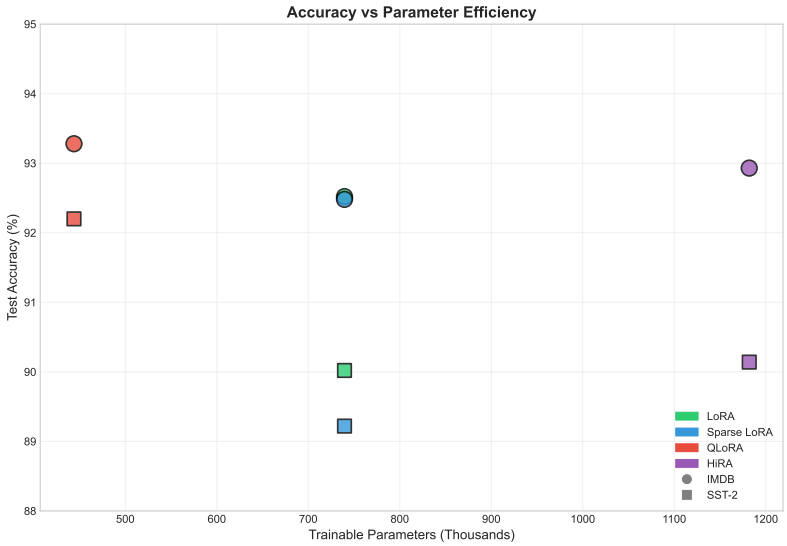


Figure 4: Accuracy vs. parameter efficiency (upper-left is better). Circles denote IMDB, squares denote SST-2.

Capacity and sparsity. Two key insights emerge from our results. First, *adapter rank matters for generation*: High-Rank LoRA's $r = 32$ adapters achieve the best perplexity without increasing training time, indicating that language modeling benefits from higher-capacity updates even when classification does not. Standard LoRA's $r = 8$ appears sufficient for sentiment tasks but underfits the more complex token prediction objective. Second, *low-rank adapters contain redundancy*: Sparse LoRA prunes 50% of adapter weights (Figure 5) with negligible accuracy loss (<0.8 percentage points), suggesting that sparse inference kernels could halve adapter compute at deployment.

Multi-metric view. The radar chart in Figure 6 summarizes the trade space: QLoRA dominates parameter and memory efficiency, Sparse LoRA leads on speed, and HiRA excels at expressiveness.

Recommendations.

- **Memory-constrained classification**: Pick QLoRA and accept the 3× longer training time in exchange for the best accuracy and 50% memory savings.

Figure 5: Sparse LoRA reaches 50% sparsity (left) while staying within 0.8 percentage points of dense LoRA on classification (right).

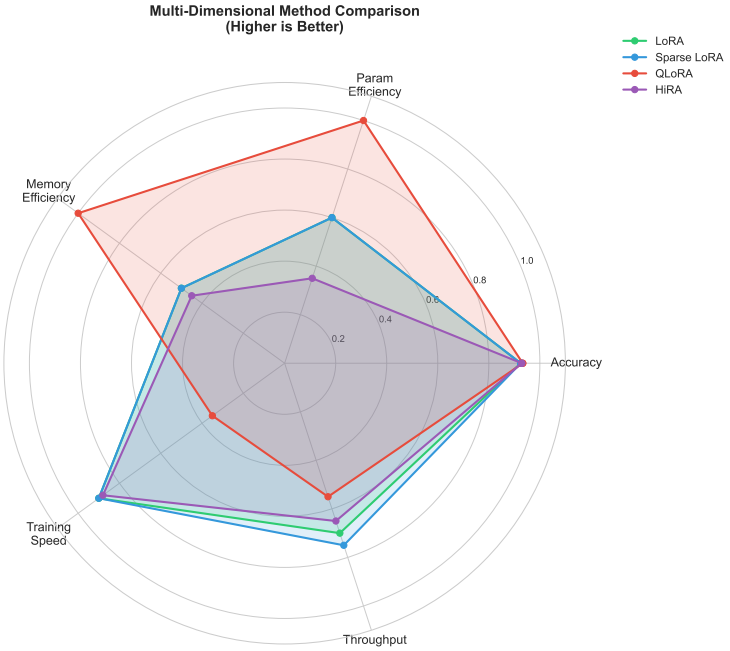


Figure 6: Multi-dimensional comparison across accuracy, parameter efficiency, memory, training speed, and throughput.

- **Fast iteration or hyper-parameter sweeps**: Use standard LoRA or HiRA; both finish in 13–35 minutes while remaining within 0.8 percentage points of QLoRA.
- **Deployment/inference efficiency**: Sparse LoRA keeps accuracy intact, opens the door to sparse kernels, and matches LoRA's training cost. Figure 7 shows LoRA-style adapters also retain the best throughput.
- **Text generation**: Prefer HiRA for superior perplexity; avoid QLoRA unless memory is the overriding constraint.

Limitations. QLoRA requires BERT-base, so parameter percentages are not directly comparable to DistilBERT-based adapters. Our experiments cover sentiment analysis and language modeling only; other tasks (QA, summarization, dialogue) may yield different conclusions. Sparse LoRA speedups assume availability of sparse kernels, and our High-Rank LoRA implementation approximates rather than replicates the full HiRA Hadamard design.

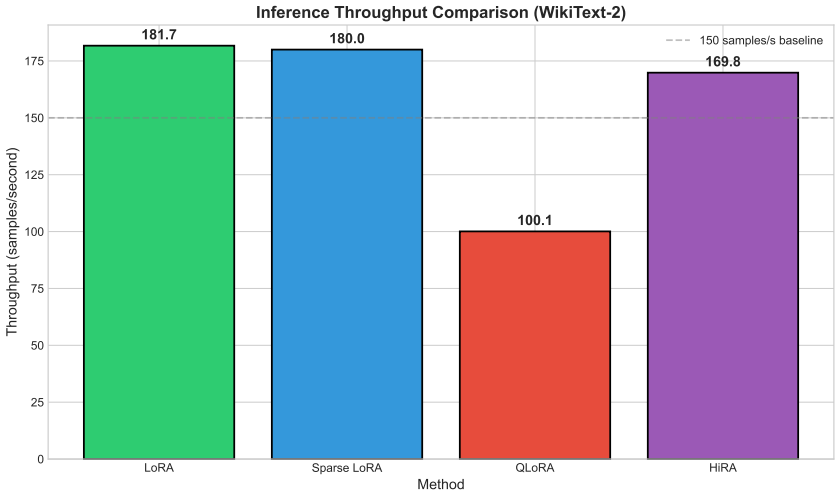


Figure 7: Inference throughput on WikiText-2. Sparse LoRA mirrors LoRA, while QLoRA drops to 100 samples/s because of dequantization overhead.

6 Conclusion

This study benchmarks the trade-offs among accuracy, speed, and memory across four PEFT strategies. We find that no single method is universally optimal: QLoRA maximizes classification accuracy through efficient memory usage, enabling larger base models, but suffers from slow training; High-Rank LoRA is critical for generative quality; and Sparse LoRA offers a path to efficient inference. For general-purpose applications, standard LoRA remains the most balanced choice. Future work should explore hybrid approaches—such as combining quantization with sparsity—and investigate scaling laws of these methods on 7B+ parameter models.

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